

Problem-Solution Fit

Date	18 July 2026
Team ID	SWTID-2026-5023
Project Name	Rising Waters (Flood Prediction System)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] - Residents in flood-prone areas - Local Government & Disaster Management Authorities - Farmers - Emergency Response Teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Poor internet connectivity - Limited smartphones - Low awareness - Power outages during floods	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: - Weather forecast websites - Government SMS alerts - News channels - Manual monitoring Limitations: - Alerts are delayed - Not location-specific - Limited prediction accuracy
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Need early flood warnings - Difficulty tracking water levels in real time - Delayed evacuation causes loss of life and property - Lack of accurate flood prediction	9. PROBLEM ROOT / CAUSE [RC] - Extreme rainfall and climate change. - Poor drainage systems. - Rising river water levels - Slow communication of warnings	7. BEHAVIOR + ITS INTENSITY [BE] - Check weather apps/news during rain - Follow social media updates - Contact local authorities - Share warnings with family
3. TRIGGERS TO ACT [TR]	10. YOUR SOLUTION [SL]	8. CHANNELS OF BEHAVIOR [CH]

- Heavy rainfall alerts
- Rapid rise in river/water levels
- Weather department warnings
- Monsoon season

4. EMOTIONS

BEFORE / AFTER [EM]

Before:

- Fear
- Anxiety
- Confusion
- Unprepared

After:

- Safe
- Confident
- Better prepared
- Reduced panic

- Predicts flood risk using rainfall and water-level data
- Sends real-time alerts via Mobile App and SMS
- Displays flood risk on an interactive dashboard
- Helps authorities and citizens prepare for evacuation
- Reduces loss of life and property

ONLINE

- Mobile App
- SMS
- WhatsApp
- Website

OFFLINE

- Sirens
- Community announcements
- Local disaster management offices
- Radio